

MamaBearZKP artifact — measured tables

Every table below is generated from the files under `results/` by `plot_tables.py`, and is **numbered and ordered as in the paper**: Table 6 here is Table 6 there. Paper Tables 2, 3, 7 and 8 are descriptive or analytically derived and correspond to no benchmark run, so the numbering skips them. Each table starts a new page, in paper order.

Table 1: Representative end-to-end prover times. Baseline is Goldilocks; both “This work” columns are MamaBear $\mathbb{F}_{p^3}$ with a 5-level FFT split. Rows are $\mu \in \{19, 21, 23\}$ of Table ??.

μ	Baseline	This work	Speed-up	This work Parallel	Speed-up
19	4.34 s	0.26 s	17x	97.3 ms	45x
21	18.61 s	1.14 s	16x	0.44 s	42x
23	75.49 s	6.37 s	12x	1.98 s	38x

Table 4: ZeroCheck and ProductCheck time across circuit sizes 2^μ for $\mu = 18$ to 28. Goldilocks and BabyBear serve as baselines. “M. vs. G./B.” reports the MamaBear speedup over Goldilocks / BabyBear. “-Par” columns use 8 threads. “-” indicates data not reported due to out-of-memory (OOM).

(a) ZeroCheck time over $\mathbb{F}_{p^3}$ with $\ell_0=2$ (§??).							(b) ProductCheck time using the per-wire design (§??) over $\mathbb{F}_{p^3}$.						
μ	Gold. $\mathbb{F}_{p^2}$	Baby. $\mathbb{F}_{p^4}$	MamaBear $\mathbb{F}_{p^3}$		MamaBear-Par $\mathbb{F}_{p^3}$		μ	Gold. $\mathbb{F}_{p^2}$	Baby. $\mathbb{F}_{p^4}$	MamaBear $\mathbb{F}_{p^3}$		MamaBear-Par $\mathbb{F}_{p^3}$	
			Time	M. vs. G./B.	Time	M. vs. G./B.				Time	M. vs. G./B.	Time	M. vs. G./B.
18	0.23 s	0.20 s	5.5 ms	42x/36x	4.1 ms	56x/49x	18	0.58 s	0.49 s	27.8 ms	21x/18x	18.3 ms	32x/27x
19	0.46 s	0.40 s	11.6 ms	40x/34x	7.4 ms	62x/54x	19	1.14 s	1.01 s	35.6 ms	32x/28x	24.3 ms	47x/42x
20	0.91 s	0.80 s	23.8 ms	38x/34x	14.3 ms	64x/56x	20	2.33 s	2.01 s	71.1 ms	33x/28x	63.3 ms	37x/32x
21	1.64 s	1.41 s	48.3 ms	34x/29x	31.1 ms	53x/45x	21	4.66 s	4.17 s	0.24 s	19x/17x	0.16 s	29x/26x
22	3.36 s	2.81 s	0.12 s	28x/23x	83.8 ms	40x/34x	22	9.50 s	8.24 s	0.52 s	18x/16x	0.34 s	28x/24x
23	6.56 s	5.66 s	0.24 s	27x/24x	0.17 s	39x/33x	23	18.64 s	15.91 s	1.12 s	17x/14x	0.69 s	27x/23x
24	13.27 s	11.46 s	0.46 s	29x/25x	0.32 s	41x/36x	24	38.41 s	34.07 s	2.19 s	18x/16x	1.32 s	29x/26x
25	26.41 s	22.52 s	0.98 s	27x/23x	0.57 s	46x/40x	25	77.30 s	67.21 s	4.35 s	18x/15x	2.57 s	30x/26x
26	52.74 s	45.72 s	1.93 s	27x/24x	1.12 s	47x/41x	26	149.8 s	132.5 s	8.88 s	17x/15x	5.12 s	29x/26x
27	105.8 s	91.00 s	3.82 s	28x/24x	2.30 s	46x/40x	27	304.1 s	266.0 s	17.68 s	17x/15x	10.16 s	30x/26x
28	210.8 s	183.8 s	7.71 s	27x/24x	4.46 s	47x/41x	28	-	-	35.13 s	-/-	20.42 s	-/-

Table 5: DeepFold performance across circuit sizes 2^μ for $\mu = 18$ to 27. Goldilocks and BabyBear serve as baselines. “M. vs. G./B.” reports the MamaBear speedup over Goldilocks / BabyBear. “-Par” columns use 8 threads. Our MamaBear implementations use 5-level splitting FFT. “-” indicates data not reported due to out-of-memory (OOM).

(a) DeepFold-Commit time.							(b) DeepFold-Open time.						
μ	Gold. $\mathbb{F}_{p^2}$	Baby. $\mathbb{F}_{p^4}$	MamaBear $\mathbb{F}_{p^3}$		MamaBear-Par $\mathbb{F}_{p^3}$		μ	Gold. $\mathbb{F}_{p^2}$	Baby. $\mathbb{F}_{p^4}$	MamaBear $\mathbb{F}_{p^3}$		MamaBear-Par $\mathbb{F}_{p^3}$	
			Time	M. vs. G./B.	Time	M. vs. G./B.				Time	M. vs. G./B.	Time	M. vs. G./B.
18	0.65 s	0.64 s	45.1 ms	14x/14x	8.6 ms	76x/74x	18	0.59 s	0.52 s	31.5 ms	19x/17x	17.7 ms	33x/29x
19	1.46 s	1.37 s	0.10 s	15x/14x	18.1 ms	81x/76x	19	1.27 s	1.10 s	63.9 ms	20x/17x	34.5 ms	37x/32x
20	2.93 s	2.83 s	0.23 s	13x/12x	40.2 ms	73x/70x	20	2.70 s	2.47 s	0.13 s	21x/19x	68.8 ms	39x/36x
21	6.08 s	5.84 s	0.51 s	12x/11x	0.10 s	61x/58x	21	5.47 s	4.96 s	0.27 s	20x/18x	0.13 s	42x/38x
22	12.23 s	11.94 s	1.19 s	10x/10x	0.23 s	53x/52x	22	11.20 s	10.21 s	0.58 s	19x/18x	0.25 s	45x/41x
23	24.82 s	24.38 s	3.19 s	8x/8x	0.53 s	47x/46x	23	22.49 s	20.46 s	1.17 s	19x/17x	0.50 s	45x/41x
24	50.32 s	49.80 s	7.50 s	7x/7x	1.18 s	43x/42x	24	45.21 s	40.88 s	2.54 s	18x/16x	1.01 s	45x/40x
25	-	-	17.01 s	-/-	2.95 s	-/-	25	-	-	5.39 s	-/-	2.10 s	-/-
26	-	-	40.03 s	-/-	7.18 s	-/-	26	-	-	11.11 s	-/-	4.16 s	-/-
27	-	-	92.34 s	-/-	17.81 s	-/-	27	-	-	22.38 s	-/-	8.37 s	-/-

Table 6: HyperPlonk–DeepFold prover time across circuit sizes 2^μ for $\mu = 18$ to 27. Goldilocks, BabyBear, and Plonky3 serve as baselines. “M. vs. G./B./P.” reports the MamaBear speedup over Goldilocks / BabyBear / Plonky3. “-Par” columns use 8 threads. Our MamaBear implementations use 5-level splitting FFT. “-” indicates data not reported due to out-of-memory (OOM).

μ	Goldilocks	BabyBear	Plonky3	Plonky3-Par	MamaBear $\mathbb{F}_{p^3}$		MamaBear-Par $\mathbb{F}_{p^3}$	
	$\mathbb{F}_{p^2}$	$\mathbb{F}_{p^4}$	B. $\mathbb{F}_{p^4}$	B. $\mathbb{F}_{p^4}$	Time	M. vs. G./B./P.	Time	M. vs. G./B./P.
18	2.07 s	1.82 s	1.82 s	0.33 s	0.10 s	21x/18x/18x	49.1 ms	42x/37x/7x
19	4.34 s	3.88 s	3.66 s	0.67 s	0.26 s	17x/15x/14x	97.3 ms	45x/40x/7x
20	8.84 s	8.07 s	7.47 s	1.35 s	0.52 s	17x/16x/14x	0.20 s	44x/40x/7x
21	18.61 s	16.60 s	15.48 s	2.64 s	1.14 s	16x/15x/14x	0.44 s	42x/38x/6x
22	37.30 s	33.86 s	30.27 s	5.84 s	2.72 s	14x/12x/11x	0.96 s	39x/35x/6x
23	75.49 s	68.64 s	61.86 s	11.82 s	6.37 s	12x/11x/10x	1.98 s	38x/35x/6x
24	152.8 s	138.0 s	128.3 s	27.20 s	14.37 s	11x/10x/9x	4.14 s	37x/33x/7x
25	–	–	–	–	31.76 s	–/–/–	8.71 s	–/–/–
26	–	–	–	–	68.77 s	–/–/–	19.10 s	–/–/–
27	–	–	–	–	144.1 s	–/–/–	–	–/–/–

Table 9: Field arithmetic microbenchmark: time (ms) for 10^8 operations on the $\mathbb{F}_{p^d}$ extension used by each backend (configuration matches Table ??). For our MamaBear addition we report the `lazy_add()` variant from §??. “M. vs. G./B./P.” is the MamaBear speedup over Goldilocks / BabyBear / Plonky3.

Op	Gold.	Baby.	Plonky3	MamaBear	M. vs. G./B./P.
Add	52.88	21.65	5.75	2.27	23.3x/9.5x/2.5x
Mul	367.40	341.00	82.78	67.92	5.4x/5.0x/1.2x
reduce_fast	–	–	–	6.87	–

Table 10: Representative circuit sizes.

Primitive	Operation	Circuit Size
SHA256	1 block	$2^{18.20}$
AES128	1 call	$2^{16.42}$
BLAKE3	1 permutation	$2^{17.19}$
Keccak-f[1600]	1 permutation	$2^{19.16}$
Poseidon2	1 permutation	$2^{12.38}$

Table 11: HyperPlonk–DeepFold proof size (in KiB) across circuit sizes 2^μ for $\mu = 18$ to 27. The MamaBear configuration uses the 5-level FFT split described in §??. “-” indicates data not reported due to out-of-memory (OOM).

μ	Goldilocks	BabyBear	Plonky3	MamaBear	M. vs. G.
18	395.11	389.43	722.88	442.87	+12.1%
19	447.72	441.07	793.58	475.82	+6.3%
20	498.41	489.98	867.83	514.99	+3.3%
21	565.80	552.34	945.07	549.26	-2.9%
22	620.20	624.84	1025.42	600.92	-3.1%
23	687.34	685.85	1109.15	654.54	-4.8%
24	756.78	753.73	1196.36	688.51	-9.0%
25	–	–	–	747.54	–
26	–	–	–	792.62	–
27	–	–	–	848.94	–

Table 12: HyperPlonk–DeepFold prover time across circuit sizes 2^μ for $\mu = 18$ to 27. Goldilocks, BabyBear, and Plonky3 serve as baselines. “M. vs. G./B./P.” reports the MamaBear speedup over Goldilocks / BabyBear / Plonky3. “-Par” columns use 8 threads. Our MamaBear implementations use 3-level splitting FFT. “-” indicates data not reported due to out-of-memory (OOM).

μ	Goldilocks	BabyBear	Plonky3	Plonky3-Par	MamaBear $\mathbb{F}_{p^3}$		MamaBear-Par $\mathbb{F}_{p^3}$	
	$\mathbb{F}_{p^2}$	$\mathbb{F}_{p^4}$	B. $\mathbb{F}_{p^4}$	B. $\mathbb{F}_{p^4}$	Time	M. vs. G./B./P.	Time	M. vs. G./B./P.
18	2.07 s	1.82 s	1.82 s	0.33 s	0.13 s	16x/14x/14x	53.1 ms	39x/34x/6x
19	4.34 s	3.88 s	3.66 s	0.67 s	0.28 s	16x/14x/13x	0.11 s	39x/35x/6x
20	8.84 s	8.07 s	7.47 s	1.35 s	0.64 s	14x/13x/12x	0.23 s	38x/35x/6x
21	18.61 s	16.60 s	15.48 s	2.64 s	1.36 s	14x/12x/11x	0.52 s	36x/32x/5x
22	37.30 s	33.86 s	30.27 s	5.84 s	3.51 s	11x/10x/9x	1.11 s	34x/31x/5x
23	75.49 s	68.64 s	61.86 s	11.82 s	7.96 s	9x/9x/8x	2.36 s	32x/29x/5x
24	152.8 s	138.0 s	128.3 s	27.20 s	17.25 s	9x/8x/7x	5.09 s	30x/27x/5x
25	–	–	–	–	37.68 s	–/–/–	11.41 s	–/–/–
26	–	–	–	–	81.48 s	–/–/–	24.05 s	–/–/–
27	–	–	–	–	–	–/–/–	–	–/–/–

Table 13: HyperPlonk–DeepFold proof size (in KiB) across circuit sizes 2^μ for $\mu = 18$ to 27. The MamaBear configuration uses the 3-level FFT split described in §??. “-” indicates data not reported due to out-of-memory (OOM).

μ	Goldilocks	BabyBear	Plonky3	MamaBear	M. vs. G.
18	395.11	389.43	722.88	312.13	-21.0%
19	447.72	441.07	793.58	344.33	-23.1%
20	498.41	489.98	867.83	388.73	-22.0%
21	565.80	552.34	945.07	428.80	-24.2%
22	620.20	624.84	1025.42	484.25	-21.9%
23	687.34	685.85	1109.15	529.12	-23.0%
24	756.78	753.73	1196.36	590.07	-22.0%
25	–	–	–	645.42	–
26	–	–	–	699.30	–
27	–	–	–	774.31	–

Table 14: HyperPlonk–DeepFold prover peak memory (in GiB) across circuit sizes 2^μ for $\mu = 18$ to 27. “M. vs. G./B./P.” reports the MamaBear memory reduction factor over Goldilocks / BabyBear / Plonky3. The MamaBear configuration uses the 5-level FFT split described in §??. “-” indicates data not reported due to out-of-memory (OOM).

μ	Gold.	Baby.	Plonky3	MamaBear		MamaBear-Par	
				Mem.	M. vs. G./B./P.	Mem.	M. vs. G./B./P.
18	0.56	0.52	0.72	0.13	4x/4x/6x	0.17	3x/3x/4x
19	1.11	1.03	1.44	0.25	4x/4x/6x	0.33	3x/3x/4x
20	2.23	2.06	2.87	0.50	4x/4x/6x	0.66	3x/3x/4x
21	4.45	4.12	5.74	1.00	4x/4x/6x	1.32	3x/3x/4x
22	8.91	8.24	11.47	2.00	4x/4x/6x	2.66	3x/3x/4x
23	17.81	16.47	22.94	4.00	4x/4x/6x	5.31	3x/3x/4x
24	35.63	32.94	45.88	8.00	4x/4x/6x	10.63	3x/3x/4x
25	–	–	–	16.00	–/–/–	21.25	–/–/–
26	–	–	–	32.00	–/–/–	42.50	–/–/–
27	–	–	–	64.00	–/–/–	85.00	–/–/–